

Pharmacokinetics and Absolute Bioavailability of Mepolizumab Following Administration at Subcutaneous and Intramuscular Sites

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Abstract

This study characterized the pharmacokinetics (PK) of mepolizumab, after a single intravenous (IV), subcutaneous (SC), or intramuscular (IM) dose in healthy adults and determined the absolute bioavailability of SC and IM mepolizumab delivered at different anatomical regions. Sixty healthy subjects were randomly assigned to receive a single dose of either mepolizumab 250 mg by IV, SC injection (upper arm, abdomen, or thigh); or IM injection (thigh). Following IV administration, the mean maximum observed plasma mepolizumab concentration (C_{max}) and the mean area under the concentration versus time curves from time zero to infinity ($AUC_{(0-\infty)}$) were $109 \pm 17 \mu\text{g/mL}$ and $1,557 \pm 250 \mu\text{g d/mL}$, respectively. After SC administration, the mean ($\pm$ SD) values of C_{max} and $AUC_{(0-\infty)}$ were $34.1\text{--}38.2 \pm 7.3\text{--}12.1 \mu\text{g/mL}$ and $1,110\text{--}1,238 \pm 228\text{--}372 \mu\text{g d/mL}$, respectively. Following IM administration, the mean values of C_{max} and $AUC_{(0-\infty)}$ were $46.9 \pm 10.6 \mu\text{g/mL}$ and $1,395 \pm 348 \mu\text{g d/mL}$. The median terminal half-life was similar for SC, IM and IV administration (17.9–20.4, 19.2, and 18.5 days, respectively). The overall mean bioavailability of SC mepolizumab was 64–75%, and absorption was relatively similar for the three SC injection sites. Mepolizumab 250 mg was generally well tolerated in this study. These results support flexibility in the SC injection site for mepolizumab.

Keywords

human anti-IL5 monoclonal antibody, mepolizumab, biologic, bioavailability, pharmacokinetics, intravenous, intramuscular, subcutaneous, absorption

In asthma, various mediators released from activated inflammatory and immune cells produce pulmonary edema, bronchoconstriction, and mucus hypersecretion that lead to changes in airway morphology.¹ It is this latter inflammatory response, mediated through recruitment of eosinophils to the lung, that is believed to be the major cause of the smooth muscle hypertrophy and chronic mucosal damage that leads to airway hyperreactivity and deterioration in lung function over time.² Eosinophil levels and interleukin (IL)-5 play an important role in maintaining the chronic inflammatory state in asthmatics. Reduction of eosinophils by blocking IL-5 activity has been shown to be of therapeutic benefit in severe asthma patients with elevated eosinophils.^{3–5}

Mepolizumab is a high-affinity, fully humanized monoclonal antibody (IgG1, kappa, mAb) specific for human IL-5. This mAb blocks binding of human IL-5 (hIL-5) to the alpha chain of the IL-5 receptor complex expressed on the eosinophil cell surface. The overall pharmacokinetic (PK) and pharmacodynamic (PD) characteristics of mepolizumab have been previously

reported.⁶ Studies in the severe asthma population have shown that more than half of these patients have persistent eosinophilic ($>3\%$ eosinophils in sputum) airway inflammation despite corticosteroid therapy.⁷

Differences in mepolizumab subcutaneous (SC) bioavailability at various injection sites may have an effect on clinical response. The intramuscular (IM) route was investigated as an alternative route of administration. The aim of the present study was to investigate the safety and PK and absolute bioavailability of mepolizumab

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following a single 250 mg intravenous (IV), SC or IM dose in healthy subjects and to compare the absolute bioavailability of three common SC injection sites (upper arm, abdomen, and thigh).

Methods

Study Design

This was a randomized, open-label, parallel-design, phase I study. Sixty healthy subjects between 18 and 55 years of age were enrolled (23 were males and 37 females). All weighed within 25% of their ideal weight. Twelve subjects were included per group and randomized using a 1:1:1:1:1 ratio. Mepolizumab was supplied as a lyophilized powder in vials containing 250 mg. Vial contents were reconstituted with sterile water for injection (WFI). For the intravenous infusion 5 mL of WFI was used to reconstitute the contents of one vial and the solution added to the 100 mL saline bag and infused over 30 minutes. Each 250 mg subcutaneous dose was administered as 2×125 mg (2×1.2 mL) injections, and the IM dose was administered as 1×250 mg (1×2.4 mL) injection.

Excluded subjects included the following: significant abnormality on clinical examination including chemistry, hematology or 12 lead ECG; history or presence of gastrointestinal, hepatic or renal disease or other condition known to interfere with the absorption, distribution, metabolism or excretion of drugs; positive Hepatitis B or Hepatitis C surface antigen within 3 months of the start of the study, exposure to more than three new chemical entities within 12 months prior to the first dosing day, or participation in a trial with any drug within 84 days before the start of the study (new chemical entity within 112 days); received a prescribed medication within 7 days prior to the first dosing day, which in the opinion of the Principal Investigator would interfere with the study procedures or compromise safety or received an over-the-counter (OTC) medicine within 48 hours before the first study day; smoking history of more than 10 pack years or history of alcohol abuse, history or presence of allergy to the study drug, or a positive pre-study urine drug screen. Women of childbearing potential were recommended to use effective contraceptive measures, oral contraceptives for at least 3 months prior to screening and continue during the study.

Subjects received a single dose of mepolizumab 250 mg on Day 1 and did consume food and fluids from midnight before the dose of mepolizumab until 1 hour post-dose (caffeine from midnight before the dose up to 24 hours post-dose). Vigorous exercise and over the counter use of medications was not permitted for 48 hours prior to screening or prior to any study visit was not permitted, and taking any prescribed medication for a period of 7 days leading to a visit was not permitted.

Study GSK 240563/018 was carried out in the GlaxoSmithKline Clinical Pharmacology Unit, Clinical Pharmacology and Discovery Medicine, Harlow, Essex, UK. The study was conducted in accordance with Good Clinical Practice guidelines, and the Declaration of Helsinki. The protocol and informed consent were approved by an Institutional Review Board/Ethics Committee prior to the start of the study. All participants provided written informed consent prior participation.

Investigational Product Administration

Each subject received one dose of mepolizumab and was followed for 12 weeks post-dose. If any subject had peripheral blood eosinophil levels below the reference range at 12 weeks after dosing they were followed until the level returned within the reference range. Each 250 mg dose was administered as two 125 mg SC injections or one 250 mg IM injection or an IV infusion. The SC dose to the arm was given as 2×1.2 mL injections into the upper outer aspect of the same arm and the SC dose to the thigh was given as two injections into the upper outer aspect of the same thigh. The SC dose to the abdomen was given as 2×1.2 mL injections into the antero-lateral wall avoiding the periumbilical region, with each injection either side of it. The IM dose was given as one 2.4 mL injection into the muscle of the lateral thigh. A narrow gauge needle was used to draw up the solution for injection to avoid particulate matter from the stopper or vial and this was replaced by a 40 mm long, 21-gauge needle that was used for the injection. The IV dose was administered in the forearm vein.

Pharmacokinetic Assessment

Serial blood samples were collected pre-dose and up to 12 weeks following SC and IM dosing, blood samples (3 mL) for PK evaluation were drawn prior to dosing, at 2, 4, 6, and 8 hours post-dose, and then on study days: 2, 3, 4, 5, 6, 8 (Week 1) and at 2, 3, 4, 6, 8, and 12 weeks post-dose. Following IV infusion, blood samples (3 mL) for PK evaluation were drawn prior to start of the 30 minutes infusion (time zero) and at the following times: after the start of infusion on Day 1 and at 15, 30 (just before the end of infusion) and 35 minutes. Further sampling was the same as indicated above for the SC and IM dosing groups.

Blood samples for PK analysis were collected into heparinized tubes and immediately chilled on crushed water-ice. Plasma was separated by centrifugation (1,500 g for 10 minutes at 4°C) and transferred into appropriately labeled polypropylene tubes, and frozen. Plasma was frozen and stored at approximately -70°C until time of shipment in the frozen state to Drug Analysis, DMPK, GSK. The samples were subsequently stored at approximately -20°C .

Bioanalytical Methods

Plasma samples were assayed for mepolizumab using a fluorescent immunoassay method (lower limit of quantification (LLQ), 0.05 µg/mL for a 0.1 mL aliquot of 10% plasma). Briefly, recombinant human IL-5 (1.214 mg/mL) was diluted in 100 mM NaHCO₃, pH 9.2 for a final concentration of 2 µg/mL. One hundred microliter of this solution was added to each well of the microtiter plate with aseptic sealing film and incubated at 4°C overnight. On the day of the analysis the microtiter plate was washed with buffer and blocked by adding 200 µL of TBS buffer containing 3% BSA. The plate was sealed and incubated at 37°C for 1 hour with constant vortexing. After additional washes and incubation the plate was sealed and incubated further at room temperature for 10 minutes with constant vortexing. The fluorescence of each well was determined using the VICTORTM multi-label counter (EG&G Wallac, Gaithersburg, MD) in the europium mode. Validation sample of mepolizumab was determined by comparison with a calibration curve prepared on the same day of the analysis and included in each run. Ninety-two ex vivo plasma samples were analyzed side-by-side by the previously validated electrochemiluminescent method⁸ and the described time-resolved fluorescent immunoassay.

Quality control (QC) samples were analyzed with each batch of experimental samples against separately prepared calibration standards. The results of the QC sample analysis were used to assess the day-to-day performance of the assay. The assay performed acceptably in a three run validation on the following concentrations: 50, 200, 1,000, 2,000, and 4,500 ng/mL. The percent average within-run precision were 6%, 4%, 3.5%, 4%, and 6.3%, respectively. The percent between-run precision were 4.5%, 2%, 1.5%, 0.5% and 1.3%, respectively. There was no significant quantifiable assay response from drug-free human plasma.

PK parameters for mepolizumab were determined from the plasma concentration versus time curves, using the actual collection times recorded at each sampling occasion. WinNonlin Professional version 2.1, (Pharsight Corporation, Mountain View, California) was used to create a non-compartmental model and calculate $C_{\max}$, the time taken to reach $C_{\max}$ ($T_{\max}$), $AUC_{(0-t)}$, $AUC_{(0-\infty)}$ and $T_{1/2}$. $AUC_{(0-\infty)}$ was estimated as the sum of $AUC_{(0-t)}$ and $C_{(t)}$ divided by the elimination rate constant, where $C_{(t)}$ was the concentration from the log-linear regression analysis at the last measurable time point. The terminal elimination rate constant (λ_z) was derived from the log-linear disposition phase of the concentration-time curve using least squares regression analysis with visual inspection of the data to determine the appropriate number of terminal data points for regression analysis (a minimum of three data points were used). $T_{1/2}$ - was calculated as $\ln 2/\lambda_z$. Mean plasma concentrations were determined from individual samples at each time point

within each regimen. Post-dose non-quantifiable values were replaced with zero for the calculation of mean concentrations.

Safety Assessments

Safety parameters included measurements of vital signs (systolic and diastolic BP); ECG intervals (PR, QRS, QT, QTc) and heart rate; hematology, clinical chemistry, and urinalysis. Subjects were closely monitored for symptoms of an injection site reaction after the administration of mepolizumab. The SC injection sites were assessed at 10 minutes, 2 hours, and 24 hours. Subjects were asked to comment on any sensations associated with the SC injections and the diameter of any visible erythema (in two directions at right angles) was measured. Any change in the injection sites such as reddening and erythema, was captured. A 5 point scale was used to assess the tolerability of the injections for the subjects: 0 = no erythema; 1 = slight erythema; 2 = well defined erythema; 3 = moderate to severe erythema; 4 = severe erythema. Observations continued if notable inflammation was still present.

Statistical Analysis

Bioavailability was estimated from the three different subcutaneous sites and the intramuscular site relative to the intravenous dose. The statistical analysis estimated the difference in systemic exposure to mepolizumab as measured by $AUC_{(0-\infty)}$ following dosing at each of the three SC sites and one IM site, relative to values obtained after IV dosing. An analysis of variance (ANOVA) model was used. The difference in $C_{\max}$ and $T_{\max}$ was also estimated. It was planned to recruit 70 subjects into the study, with fourteen subjects in each of the five treatment groups to ensure at least 12 evaluable subjects per group. Based on a between-subject standard deviation of 0.35 for log $AUC_{(0-\infty)}$ (following a single 250 mg SC dose in a previous study GSK 240563/017), it was estimated that the lower and upper bounds of the 90% confidence intervals for the ratios "SC:IV" and "IM:IV" would be within 28% of the point estimate for $AUC_{(0-\infty)}$. Primary analysis was $AUC_{(0-\infty)}$ of mepolizumab. A secondary analysis included $C_{\max}$, $T_{\max}$, and $T_{1/2}$ of mepolizumab. AUC and $C_{\max}$ were log transformed (base e) and analyzed separately by ANOVA, fitting a model which included variables for regimen, weight and gender. $T_{\max}$ was analyzed non-parametrically using the Mann-Whitney method. Point estimates for the ratio of each regimen compared to the IV route of administration and their related 90% confidence interval (CI) were calculated.

Results

Study Subjects

Sixty healthy subjects between the ages of 20–52 years were recruited (12 in each group). Twenty-three were

males and 37 females, 57 were white, and 3 of mixed race “other,” and all weighed within 25% of their ideal weight (range 51.7–96.1 kg). All randomized subjects completed the study. Demographic characteristics were comparable with respect to age, race, weight, and height across treatment groups.

Pharmacokinetics

Following the 30 minutes IV infusion, mepolizumab plasma concentrations declined in a seemingly bi-exponential manner. T_{max} ranged from 0.5 to 4 hours, although plasma concentrations remained relatively constant during this time period. Following administration of the SC injections at three different sites (upper arm, abdomen, and thigh) and the IM injection, the mean mepolizumab plasma concentration-time profiles were similar in shape (Figure 1). The mean plasma concentrations were higher following IM administration. Mepolizumab was absorbed slowly following SC and IM administration, with T_{max} values ranging from 2 to 14 days.

The mean (90% CI) percent bioavailability for each injection site was 64 (55–73), 75 (66–86), 71 (62–82), and 81 (71–94) for SC abdomen, arm, thigh or IM, respectively. The arithmetic mean ($\pm$ SD) $AUC_{(0-\infty)}$ was 1,110 ($\pm$ 372) μ g d/mL; 1,238 ($\pm$ 228) μ g d/mL; 1,196 ($\pm$ 254) μ g d/mL; 1,395 ($\pm$ 348) μ g d/mL; and 1,557 ($\pm$ 250) μ g d/mL for SC abdomen, SC arm, SC thigh, IM and IV, respectively. For all administrations, mepolizumab was quantifiable for up to 12 weeks post-dosing. The extrapolated AUC was less than 20% of the total $AUC_{(0-\infty)}$ in all subjects, indicating that the reported

$AUC_{(0-\infty)}$ values are an accurate measure of exposure to mepolizumab in these subjects. The terminal phase half-life was long, ranging from 11 to 26 days and was independent of dosing route. For six subjects, concentration values greater than the LLQ were reported at sampling time zero. Generally, these concentrations were close to the LLQ of the drug assay and were less than 0.6% of the C_{max} values observed in these subjects. These quantifiable pre-dose concentrations were set to zero for the computation of $AUC_{(0-\infty)}$.

The between subject coefficient of variation (CV) for systemic exposure to mepolizumab in this study were generally low, with the IV route resulting in CVs of 16.4% and 15.6% for AUC and C_{max} , respectively. The SC (abdomen) route showed the highest variability in both AUC and C_{max} with values of 32.2% and 32.5%, respectively. The three other regimens have similar CVs ranging from 18.7% to 26.4% and 20.7% to 24.7%, for AUC and C_{max} , respectively. As expected, a higher C_{max} was observed following IV dosing. Dose normalized AUC and C_{max} (on a mg/kg basis) values show that generally the range of values were similar in males and females. There was no clear evidence of outliers in the analysis of either AUC or C_{max} . The ANOVA results of the fitted model for AUC and C_{max} are summarized in Table 1. The overall mean $\pm$ SD absolute bioavailability after SC and IM administration of mepolizumab was $72.8 \pm 7.1\%$.

A reduction in peripheral blood eosinophils was observed in these healthy subjects and was similar to that seen in subjects with asthma (data not shown).

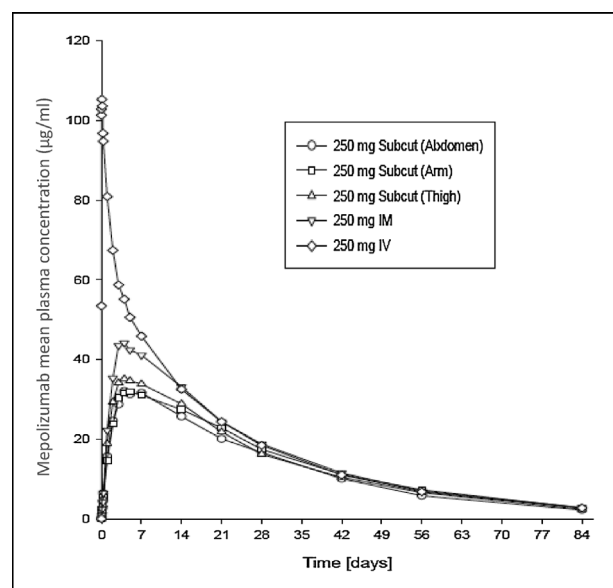


Figure 1. Mean Plasma concentration following administration of mepolizumab 250 mg via different routes with permission from Springer Science + Business Media, B.V.

Table 1. Results of the Statistical Analyses for the Comparison of $AUC_{(0-\infty)}$, C_{max} , and T_{max} for the Subcutaneous and Intramuscular Administration, Relative to Values Obtained After the Intravenous Administration

	Comparison	Point Estimate	90% CI
$AUC_{(0-\infty)}$	A:E	0.64	0.55, 0.73
	B:E	0.75	0.66, 0.86
	C:E	0.71	0.62, 0.82
	D:E	0.81	0.71, 0.94
C_{max}	A:E	0.27	0.24, 0.31
	B:E	0.29	0.26, 0.33
	C:E	0.31	0.27, 0.36
	D:E	0.38	0.33, 0.43
T_{max}	A:E	6.93	4.21, 7.09
	B:E	5.13	4.99, 6.91
	C:E	5.12	4.04, 6.86
	D:E	3.96	3.12, 5.03

A = 250 mg SC (abdomen), B = 250 mg SC (arm), C = 250 mg SC (thigh), D = 250 mg IM, E = 250 mg IV.

Assessment of Injection Sites

Sites of injection were monitored post-dose; there was no clinically important erythema and no reports of local irritation. Mepolizumab was well tolerated by all subjects irrespective of the site or route of injection and 15 injection sites assessment reported measurable wheals with most at 10 minutes post-dose, out of 144 wheal injection assessments. Not all measures of the injection sites were given erythema scores because some were pale, raised areas indicating that liquid was lying beneath the skin but not causing erythema. Erythema scores were 1 (slight erythema) except one subject (SC to abdomen) who had a score of 2 (well defined erythema) at 10 minutes but zero at subsequent times. Three subjects reported pain associated with the injections and this was only at 10 minutes post-dose. Symptoms were described as mild numbness, tingling and slight soreness. No site was scored above 2. Slight reddening of approximately 15% of the sites was visible at 10 minutes post-dose and occasionally at 2 hours post-dose. No erythema or irritation was detectable at 24 hours post-dose.

Safety

Adverse events (AEs) were typical of those reported in studies recruiting healthy subjects, with headache, pharyngitis and upper respiratory tract infections (URTI) occurring most frequently. There were no deaths or serious AEs in this study. Of the 10 AEs classified as having a probable relationship with the drug, 5 were headaches starting within 6 hours of dosing, one was dizziness starting 2 hours 5 minutes after the IV dose, 2 were purpura following SC injection to the abdomen and the arm, and 2 were pain in the leg: one following SC injection into the thigh and one following IM injection into the thigh. All were mild and all resolved within 24 hours with the exception of the purpura that resolved in 4 days 11 hours and 1 week. The majority of the AEs were classified as unrelated to study medication and mild, a few moderate and six were reported as severe.

There were no clinically important findings on examination of the vital signs and ECG data and no differences between groups, gender, or trends or changes over time. Overall, no clinically meaningful changes were observed in clinical assessments including laboratory and vital signs across the treatment groups.

Conclusions

This study was designed to determine the PK parameters of a single dose of mepolizumab 250 mg when administered as SC or IM injection or as an IV infusion in healthy subjects and to compare the absolute bioavailability of SC mepolizumab administered at different anatomical regions (upper arm, abdomen, or thigh). Monoclonal

antibodies (mAbs) have been used in the treatment of various diseases over the past two decades and combine high specificity with generally low toxicity. Monoclonal antibodies are generally administered subcutaneously, intravenously, or intramuscularly.⁹ Their PK properties differ markedly from those of non-antibody-type drugs, and these properties can have important clinical implications.

The relative bioavailabilities of mepolizumab following administration via SC administration at the different sites (abdomen, arm and thigh) were 64%, 75% and 71%, respectively, when compared to values obtained after administration by the intravenous route. The bioavailability of mepolizumab following an intramuscular injection was 81%, when compared to values obtained after administration by the IV route. The overall mean $\pm$ SD absolute bioavailability after SC and IM administration of mepolizumab 250 mg was $72.8 \pm 7.13\%$. This is consistent with other monoclonal antibodies delivered subcutaneously such as omalizumab (62%),¹⁰ golimumab (51%),¹¹ efalizumab (50%),¹² and adalimumab (64%).¹³ The basis for the 50–70% bioavailability of protein drugs after SC injection is not entirely understood, although the pre-systemic loss of proteins has been generally attributed to the degradation or catabolism at the site of injection or during the transport through the lymphatic system.¹⁴ The lymphatic system plays an important role in the absorption of subcutaneously administered proteins,¹⁵ and an increase in molecular weight results in an increased contribution of lymphatic absorption relative to the absorption through blood capillaries.^{16,17} IgG-type monoclonal antibodies have a molecular weight of approximately 150 kDa, and the lymphatic pathway is thus thought to be the primary mechanism for the absorption of monoclonal antibodies from the SC injection site.

In this context, regional variation in lymph flow is likely. However, the present study showed that the PK parameters ($C_{\max}$, $T_{\max}$, and bioavailability) of SC mepolizumab were minimally affected by injection sites (upper arm, abdomen, and thigh), although the abdomen was slightly lower in terms of $AUC_{(0-\infty)}$. It is possible that the inherent slow absorption of monoclonal antibodies makes the regional variation of lymph flow negligible. The relatively similar PK profile and absolute bioavailability of mepolizumab coupled with low CV after SC administration at different anatomical regions suggest that patients who receive SC mepolizumab treatment can have the flexibility to choose an injection location of their preference. SC dosing provides ease of administration, cost-effectiveness, and is preferred by patients and healthcare providers.¹⁸

The median $T_{1/2}$ of mepolizumab following IV administration was 18.5 days and is similar following SC and IM routes of administration ranging from 19.9 to

20.4 days, suggesting that the administration route did not affect the elimination of mepolizumab. Although the absorption of mepolizumab from the SC injection sites might be expected to result in additional variability, the variability in $AUC_{(0-\infty)}$ following SC injection (range between 18% and 26% CV) was not considerably greater than that observed with IV administration (16% CV), minimizing any concerns.

The effect of body weight in this study was investigated and consistent with a recent study⁴ in 616 subjects with severe refractory asthma where the impact of body weight on the total exposure of mepolizumab measured as area under the plasma concentration-time curve (AUC) was assessed and considered not significant to influence efficacy or safety. For example, a 30% increase in body weight would result in a 16% decrease in AUC, which falls within the between subject variability in clearance after taking into account body weight. This small change is not considered meaningful for efficacy outcomes. Overall, the parenteral administration, slow tissue distribution and long elimination half-life appear to be the most pronounced clinical PK characteristics of monoclonal antibodies.

Overall, mepolizumab was well tolerated both locally and systemically following all routes and sites of administration. This study showed that the absorption and elimination of mepolizumab was relatively similar following a single SC injection in the upper arm, abdomen, or thigh and IM injection in the thigh.

Declaration of Interest

All authors are employees of GlaxoSmithKline.

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